

T0 Theory: The Gravitational Constant

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Abstract

This document presents the systematic derivation of the gravitational constant G from the fundamental principles of T0 theory. The complete formula $G_{\text{SI}} = \frac{\xi_0^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$ explicitly shows all required conversion factors. The form and order of magnitude of G follow from ξ and one anchor; with the SI conversion factor C_{conv} the value lies +0.003% above the measured value. Since C_{conv} additionally contains the small residual of the single-anchor chain (Doc. 383), this agreement does not count as an independent precision result. Special attention is given to the physical justification of the conversion factors that establish the connection between geometric theory and measurable quantities.

1 Introduction: Gravitation in T0 Theory

The Problem of the Gravitational Constant

The gravitational constant $G = 6.674 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ is one of the least precisely known natural constants. Its theoretical derivation from first principles is one of the great unsolved problems in physics.

Key Result

T0 Hypothesis for Gravitation:

The gravitational constant is not fundamental but follows from the geometric structure of three-dimensional space through the relation:

$$G_{\text{SI}} = \frac{\xi_0^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (1)$$

where all factors are derivable from geometry or fundamental constants.

Overview of the Derivation

The T0 derivation proceeds in four systematic steps:

1. **Fundamental T0 Relation:** $\xi = 2\sqrt{G \cdot m_{\text{char}}}$
2. **Solution for G:** $G = \frac{\xi^2}{4m_{\text{char}}}$ (natural units)
3. **Dimensional Correction:** Transition to physical dimensions
4. **SI Conversion:** Conversion to experimentally comparable units

.2 The Fundamental T0 Relation

Geometric Basis

Starting Point of T0 Gravitation Theory:

T0 theory postulates a fundamental geometric relation between the characteristic length parameter ξ and the gravitational constant:

$$\xi = 2\sqrt{G \cdot m_{\text{char}}} \quad (2)$$

Geometric Interpretation: This equation describes how the characteristic length scale ξ (defined by the tetrahedral space structure) determines the strength of gravitational coupling. The factor 2 corresponds to the dual nature of mass and space in T0 theory.

Physical Interpretation:

- ξ encodes the geometric structure of space (tetrahedral packing)
- G describes the coupling between geometry and matter
- m_{char} sets the characteristic mass scale

Solution for the Gravitational Constant

Solving equation (2) for G yields:

$$G = \frac{\xi^2}{4m_{\text{char}}} \quad (3)$$

Significance: This fundamental relation shows that G is not an independent constant but is determined by space geometry (ξ) and the characteristic mass scale (m_{char}).

Choice of Characteristic Mass

T0 theory uses the electron mass as the characteristic scale:

$$m_{\text{char}} = m_e = 0.511 \text{ MeV} \quad (4)$$

The justification lies in the electron's role as the lightest charged particle and its fundamental importance for electromagnetic interaction.

.3 Dimensional Analysis in Natural Units

Unit System of T0 Theory

Dimensional Analysis

Dimensional Analysis in Natural Units:

T0 theory works in natural units with $\hbar = c = 1$:

$$[M] = [E] \quad (\text{from } E = mc^2 \text{ with } c = 1) \quad (5)$$

$$[L] = [E^{-1}] \quad (\text{from } \lambda = \hbar/p \text{ with } \hbar = 1) \quad (6)$$

$$[T] = [E^{-1}] \quad (\text{from } \omega = E/\hbar \text{ with } \hbar = 1) \quad (7)$$

The gravitational constant therefore has the dimension:

$$[G] = [M^{-1}L^3T^{-2}] = [E^{-1}][E^{-3}][E^2] = [E^{-2}] \quad (8)$$

Dimensional Consistency of the Basic Formula

Checking equation (3):

$$[G] = \frac{[\xi^2]}{[m_{\text{char}}]} \quad (9)$$

$$[E^{-2}] = \frac{[1]}{[E]} = [E^{-1}] \quad (10)$$

The basic formula is not yet dimensionally correct. This shows that additional factors are required.

.4 The First Conversion Factor: Dimensional Correction

Origin of the Correction Factor

Derivation of the Dimensional Correction Factor:

To go from $[E^{-1}]$ to $[E^{-2}]$, we need a factor with dimension $[E^{-1}]$:

$$G_{\text{nat}} = \frac{\xi_0^2}{4m_e} \times \frac{1}{E_{\text{char}}} \quad (11)$$

where E_{char} is a characteristic energy scale of T0 theory.

Determination of E_{char} :

From the derivation in the section "Derivation of the characteristic energy scale" follows:

$$E_{\text{char}} = 28.8 \quad (\text{natural units}) \quad (12)$$

This corresponds to the reciprocal of the first conversion factor:

$$C_1 = \frac{1}{E_{\text{char}}} = \frac{1}{28.8} = 3.472 \times 10^{-2} \quad (13)$$

E_{char} cancels in G (division in step 2, multiplication in step 3); its value therefore does not affect G .

Physical Significance of E_{char}

Key Result

The Characteristic T0 Energy Scale:

$E_{\text{char}} = 28.8$ (natural units) represents a fundamental intermediate scale:

$$E_0 = 7.398 \text{ MeV} \quad (\text{electromagnetic scale}) \quad (14)$$

$$E_{\text{char}} = 28.8 \quad (\text{T0 intermediate scale}) \quad (15)$$

$$E_{T0} = \frac{1}{\xi_0} = 7500 \quad (\text{fundamental T0 scale}) \quad (16)$$

This hierarchy $E_0 \ll E_{\text{char}} \ll E_{T0}$ reflects the different coupling strengths.

.5 Derivation of the Characteristic Energy Scale

Geometric Basis

The characteristic energy scale $E_{\text{char}} = 28.8 \text{ MeV}$ arises from the fundamental fractal structure of T0 theory:

$$E_{\text{char}} = E_0 \cdot R_f^2 \cdot g \cdot K_{\text{renorm}} \quad (17)$$

$$= 7.400 \times \left(\frac{4}{3}\right)^2 \times \frac{\pi}{\sqrt{2}} \times 0.986 \quad (18)$$

$$= 28.8 \text{ MeV} \quad (19)$$

Explanation of Factors:

- $E_0 = 7.400 \text{ MeV}$: Fundamental reference energy from electromagnetic scale
- $R_f = \frac{4}{3}$: Fractal scaling ratio (tetrahedral packing density)
- $g = \frac{\pi}{\sqrt{2}}$: Geometric correction factor (deviation from Euclidean geometry)
- $K_{\text{renorm}} = 0.986$: Fractal renormalization (consistent with K_{frak})

Stage 1: Fundamental Reference Energy

From the fine-structure constant derivation in T0 theory, the fundamental reference energy is known:

$$E_0 = 7.400 \text{ MeV} \quad (20)$$

This energy scales the electromagnetic coupling in T0 geometry.

Stage 2: Fractal Scaling Ratio

T0 theory postulates a fundamental fractal scaling ratio:

$$R_f = \frac{4}{3} \quad (21)$$

This ratio corresponds to the tetrahedral packing density in three-dimensional space and appears in all scaling relations of T0 theory.

Stage 3: First Resonance Stage

Application of the fractal scaling ratio to the reference energy:

$$E_1 = E_0 \cdot R_f^2 = 7.400 \times \left(\frac{4}{3}\right)^2 = 7.400 \times 1.777 \dots = 13.156 \text{ MeV} \quad (22)$$

The quadratic application (R_f^2) corresponds to the next higher resonance stage in the fractal vacuum field.

Stage 4: Geometric Correction Factor

Accounting for geometric structure through the factor:

$$g = \frac{\pi}{\sqrt{2}} \approx 2.221 \quad (23)$$

This factor describes the deviation from ideal Euclidean geometry due to the fractal spacetime structure.

Stage 5: Preliminary Value

Combination of all factors:

$$E_{\text{prelim}} = E_0 \cdot R_f^2 \cdot g = 7.400 \times 1.777 \dots \times 2.221 \approx 29.2 \text{ MeV} \quad (24)$$

Stage 6: Fractal Correction

The final correction accounts for the effective fractal dimension $D_f^{\text{eff}} \approx 2.973$ (cumulative over the RG running) with the consistent formula:

$$K_{\text{renorm}} = 1 - \frac{D_f - 2}{68} = 1 - \frac{0.973}{68} = 0.986 \quad (25)$$

Stage 7: Final Value

Application of fractal renormalization:

$$E_{\text{char}} = E_{\text{prelim}} \cdot K_{\text{renorm}} = 29.2 \times 0.986 \approx 28.8 \text{ MeV} \quad (26)$$

Consistency with the Gravitational Constant

The consistent application of the fractal correction is crucial:

- For G_{SI} : $K_{\text{frak}} = 0.986$
- For E_{char} : $K_{\text{renorm}} = 0.986$
- Same formula: $K = 1 - \frac{D_f - 2}{68}$
- Same effective fractal dimension: $D_f^{\text{eff}} \approx 2.973$

.6 Fractal Corrections

The Fractal Spacetime Dimension

Quantum Spacetime Corrections:

T0 theory accounts for the fractal structure of spacetime at Planck scales:

$$D_f^{\text{eff}} \approx 2.973 \quad (\text{effective fractal dimension over RG running}) \quad (27)$$

$$K_{\text{frak}} = 1 - \frac{D_f - 2}{68} = 1 - \frac{0.973}{68} = 0.986 \quad (28)$$

Geometric Meaning: The factor 68 corresponds to the tetrahedral symmetry of the T0 space structure. The effective fractal dimension $D_f^{\text{eff}} \approx 2.973$ describes the cumulative effect of spacetime porosity ($D_f^{\text{space}} = 3 - \xi \approx 2.9999$) over the RG running.

Physical Effect:

- Reduces gravitational coupling strength by 1.4%
- Together with C_{conv} , leads to +0.003% relative to the measured value (assessment in the section "Experimental Comparison")
- Is consistent with the renormalization of the characteristic energy

Fractal Dimension and Fine-Structure Constant**Link to the Fine-Structure Constant:**

The fractal dimension is linked to the fine-structure constant via the consistency condition of two independent derivation paths for the mass ratio m_e/m_μ (cf. Doc. 133).

Key Observation:

- The fine-structure constant can be derived **in two independent ways**:
 1. From the mass ratios of elementary particles **without fractal correction**
 2. From the fundamental T0 geometry **with fractal correction**
- Both derivations must yield the **same numerical value** for α
- This condition fixes D_f

Consistency Condition:

$$\alpha_{\text{Masses}} = \alpha_{\text{Geometry}} \times K_{\text{frak}} \quad (29)$$

$$\frac{1}{137.036} = \alpha_0 \times \left(1 - \frac{D_f - 2}{68}\right) \quad (30)$$

With $\alpha_{\text{Masses}} = \xi m_e m_\mu = 1/138.911$ and the measured value $1/137.036$ one gets $K = 137.036/138.911 = 0.98650$, hence $D_f - 2 = 68 \times 0.01350 = 0.918$, $D_f = 2.918$. The equation therefore does not yield the value $D_f^{\text{eff}} \approx 2.973$ used in this document: that value gives the factor 0.98569 and thus $137.036/0.98569 = 139.03$ or $137.036 \times 0.98569 = 135.08$ – neither is 138.911. The value 2.973 is not supported by this condition; the corpus anchor for α is $K_{\text{frak}} = 1 - 100\xi$ with $E_0 = 7.348/\sqrt{K_{\text{frak}}}$.

Physical Significance: The fractal correction $K = 1 - \frac{D_f - 2}{68}$ links

- The electromagnetic coupling (fine-structure constant)
- The gravitational coupling (gravitational constant)
- The mass scales of elementary particles within a common geometric framework.

Effect on the Gravitational Constant

The fractal correction modifies the gravitational constant:

$$G_{\text{frak}} = G_{\text{ideal}} \times K_{\text{frak}} = G_{\text{ideal}} \times 0.986 \quad (31)$$

Together with C_{conv} , this 1.4% reduction brings the theoretical prediction to within +0.003% of the measured value (assessment in the section "Experimental Comparison").

.7 The Second Conversion Factor: SI Conversion

From Natural to SI Units

Dimensional Analysis

Conversion from $[E^{-2}]$ to $[\text{m}^3/(\text{kg} \cdot \text{s}^2)]$:

The conversion proceeds via fundamental constants:

$$1 \text{ (nat. unit)}^{-2} = 1 \text{ GeV}^{-2} \quad (32)$$

$$= 1 \text{ GeV}^{-2} \times \left(\frac{\hbar c}{\text{MeV} \cdot \text{fm}} \right)^3 \times \left(\frac{\text{MeV}}{c^2 \cdot \text{kg}} \right) \times \left(\frac{1}{\hbar \cdot \text{s}^{-1}} \right)^2 \quad (33)$$

After systematic application of all conversion factors, we obtain:

$$C_{\text{conv}} = 7.783 \times 10^{-3} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \text{MeV} \quad (34)$$

Physical Significance of the Conversion Factor

The factor C_{conv} encodes the fundamental conversions:

- Length conversion: $\hbar c$ for GeV to meters
- Mass conversion: Electron rest energy to kilograms
- Time conversion: $\hbar$ for energy to frequency

.8 Numerical Verification

Step-by-Step Calculation

Verification

Detailed Numerical Evaluation:

Step 1: Calculate basic term

$$\xi_0^2 = \left(\frac{4}{3} \times 10^{-4}\right)^2 = 1.7778 \times 10^{-8} \quad (35)$$

$$\frac{\xi_0^2}{4m_e} = \frac{1.7778 \times 10^{-8}}{4 \times 0.511} = 8.698 \times 10^{-9} \text{ MeV}^{-1} \quad (36)$$

Step 2: Apply conversion factors

$$G_{\text{inter}} = 8.698 \times 10^{-9} \times 3.472 \times 10^{-2} = 3.020 \times 10^{-10} \quad (37)$$

$$G_{\text{nat}} = 3.020 \times 10^{-10} \times 7.783 \times 10^{-3} = 2.350 \times 10^{-12} \quad (38)$$

Step 3: Fractal correction and compensation of E_{char}

$$G_{\text{SI}} = 2.350 \times 10^{-12} \times 0.986 \times 28.8 \quad (39)$$

$$= 6.6745 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2} \quad (40)$$

The factor $E_{\text{char}} = 28.8$ in step 3 compensates the division by E_{char} in step 2; E_{char} cancels, and the result 6.6745×10^{-11} is the same as in the short form $\frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$.

Experimental Comparison

Verification

Comparison with Experimental Values:

Source	$G [10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}]$	Uncertainty
CODATA 2018	6.67430	± 0.00015
T0 Prediction	6.6745	(calculated)
Deviation	+0.003%	(with residual in C_{conv})

Comparison of the T0 Gravitational Formula with Experiment

Relative Precision: With $\xi_0 = 4/30000$, $m_e = 0.511 \text{ MeV}$, $C_{\text{conv}} = 7.783 \times 10^{-3}$ and $K_{\text{frak}} = 0.986$ the formula gives 6.6745×10^{-11} ; the T0 prediction agrees with experiment to about 1 part in 30,000 (+0.003%). A higher precision is not supported by the four-digit C_{conv} .

Assessment: C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383); with the anchor m_e , G lies at -2.6% , with v at -0.46% (Doc. 383). The agreement to 0.003% arises only with this residual and does not count as an independent precision result.

.9 Consistency Check of the Fractal Correction

Independence of Mass Ratios

Key Result

Consistency of Fractal Correction:

The fractal correction K_{frak} cancels out in mass ratios:

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}} \quad (41)$$

Interpretation: This explains why mass ratios can be calculated directly from fundamental geometry, while absolute mass values require the fractal correction.

Consequences for the Theory

Explanation of Observed Phenomena:

This property explains why in physics:

- **Mass ratios** can be correctly calculated without fractal correction
- **Absolute masses and coupling constants**, however, require the fractal correction
- The **fine-structure constant** α can be derived both from mass ratios (uncorrected) and from geometric principles (corrected)

Mathematical Consistency:

$$\text{Mass ratio: } \frac{m_i}{m_j} = \frac{K_{\text{frak}} \cdot m_i^{\text{bare}}}{K_{\text{frak}} \cdot m_j^{\text{bare}}} = \frac{m_i^{\text{bare}}}{m_j^{\text{bare}}} \quad (42)$$

$$\text{Absolute value: } m_i = K_{\text{frak}} \cdot m_i^{\text{bare}} \quad (43)$$

$$\text{Gravitational constant: } G = \frac{\xi_0^2}{4m_e^{\text{bare}}} \times K_{\text{frak}} \quad (44)$$

Experimental Confirmation

Verification

Verification of Theoretical Consistency:

T0 theory makes the following testable predictions:

1. **Mass ratios** can be calculated directly from fundamental geometry
2. **Absolute masses** require the fractal correction $K_{\text{frak}} = 0.986$
3. **Coupling constants** (G, α) are consistent with the same correction
4. The **effective fractal dimension** $D_f^{\text{eff}} \approx 2.973$ is universal for all scaling phenomena

Example: Muon-Electron Mass Ratio

$$\frac{m_\mu}{m_e} = \frac{12}{5} \xi^{-1/2} = 207.85 \quad (\text{T0 geometry without } K_{\text{frak}}) \quad (45)$$

lies 0.52 % above the experimental value 206.768; the ratio does not need K_{frak} because the factor cancels in it, while the absolute masses require the correction.

.10 Physical Interpretation

Meaning of the Formula Structure

Key Result

The T0 Gravitational Formula Reveals the Fundamental Structure:

$$G_{\text{SI}} = \underbrace{\frac{\xi_0^2}{4m_e}}_{\text{Geometry}} \times \underbrace{C_{\text{conv}}}_{\text{Units}} \times \underbrace{K_{\text{frak}}}_{\text{Quantum}} \quad (46)$$

1. **Geometric Core:** $\frac{\xi_0^2}{4m_e}$ represents the fundamental space-matter coupling
2. **Units Bridge:** C_{conv} connects geometric theory with measurable quantities
3. **Quantum Correction:** K_{frak} accounts for the fractal quantum spacetime

Comparison with Einsteinian Gravitation

Aspect	Einstein	T0 Theory
Basic Principle	Spacetime Curvature	Geometric Coupling
G-Status	Empirical Constant	Derived Quantity
Quantum Corrections	Not Considered	Fractal Dimension
Predictive Power	None for G	Form and order of magnitude from ξ and one anchor
Unity	Separate from QM	Unified with Particle Physics

Comparison of Gravitational Approaches

.11 Theoretical Consequences

Modifications of Newtonian Gravitation

Warning

T0 Predictions for Modified Gravitation:

T0 theory predicts deviations from Newton's law of gravitation at characteristic length scales:

$$\Phi(r) = -\frac{GM}{r} [1 + \xi_0 \cdot f(r/r_{\text{char}})] \quad (47)$$

where $r_{\text{char}} = \xi_0 \times \text{characteristic length}$ and $f(x)$ is a geometric function in the modified gravitational potential (not a frequency and not the base winding number $f = 1/\xi$).

Experimental Signature: At distances $r \sim 10^{-4} \times \text{system size}$, 0.01% deviations should be measurable.

Cosmological Implications

T0 gravitation theory has far-reaching consequences for cosmology:

1. **Dark Matter:** Could be explained by ξ_0 field effects
2. **Dark Energy:** Not required in static T0 universe
3. **Hubble Constant:** Effective expansion through redshift
4. **Big Bang:** Replaced by eternal, cyclic model

.12 Methodological Insights

Importance of Explicit Conversion Factors

Key Result

Central Insight:

The systematic treatment of conversion factors is essential for:

- Dimensional consistency between theory and experiment
- Transparent separation of physics and conventions
- Traceable connection between geometric and measurable quantities
- Precise predictions for experimental tests

This methodology should become standard for all theoretical derivations.

Significance for Theoretical Physics

The successful T0 derivation of the gravitational constant shows:

- Geometric approaches can provide quantitative predictions
- Fractal quantum corrections are physically relevant
- Unified description of gravitation and particle physics is possible
- Dimensional analysis is indispensable for precise theories